

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-4

Course: ECE/EEE F311 Communication Systems
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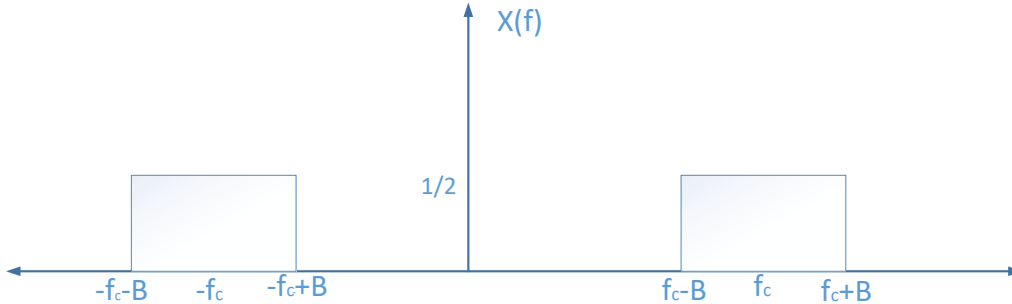
Date: 10-09-2025

Objectives

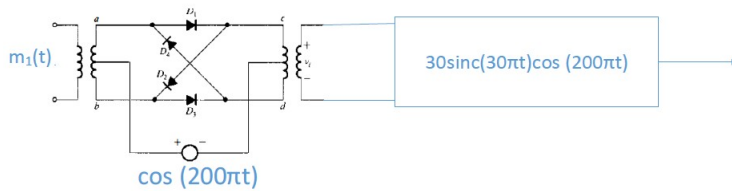
- SSB Modulation, AM Generation Circuits.

Tutorial Problems

1. A DSB-SC modulated signal at the carrier frequency f_c in the frequency domain is shown in figure below. Find the USSB in the time domain.



2. A message signal $m_1(t)$ of bandwidth 10 Hz amplitude modulates a carrier signal using the balanced ring modulator, as shown in the figure. Find the signal at the output of the bandpass filter.



Solution

1. From the DSB-SC spectrum, $m(t) = 2B \text{sinc}(2B\pi t)$
 $m(t) = \frac{\sin(2B\pi t)}{\pi t} = 2B \frac{\sin(B\pi t) \cos(B\pi t)}{B\pi t} = 2B \text{sinc}(B\pi t) \cos(B\pi t)$.
 Let $a(t)$ be *lowpass* and let $b(t)$ be *bandpass*

Then

$$H\{a(t)b(t)\} = a(t)H\{b(t)\}.$$

$$m_h(t) = \mathcal{H}\{2B \text{sinc}(B\pi t) \cos(B\pi t)\} = 2B \text{sinc}(\pi B t) \sin(\pi B t) = \frac{2 \sin^2(B\pi t)}{\pi t} = \frac{1 - \cos(2\pi B t)}{\pi t}$$

$$x_{\text{USB}}(t) = m(t) \cos(2\pi f_c t) - \hat{m}(t) \sin(2\pi f_c t) = \frac{1}{\pi t} \left[\sin(2\pi B t) \cos(2\pi f_c t) - (1 - \cos(2\pi B t)) \sin(2\pi f_c t) \right].$$

Using trigonometric identity $\sin(A + B)$ in the above

$$x_{\text{USB}}(t) = \frac{1}{\pi t} \left[\sin(2\pi(f_c + B)t) - \sin(2\pi f_c t) \right].$$

Home Work (Not to be solved): Another approach using Direct IFT:

$$S_{\text{USB}}(f) = \begin{cases} \frac{1}{2}, & f \in [f_c, f_c + B], \\ \frac{1}{2}, & f \in [-(f_c + B), -f_c], \\ 0, & \text{otherwise,} \end{cases}$$

$$\begin{aligned} s_{\text{USB}}(t) &= \int_{-\infty}^{\infty} S_{\text{USB}}(f) e^{j2\pi f t} df \\ &= 0.5 \int_{f_c}^{f_c+B} e^{j2\pi f t} df + 0.5 \int_{-(f_c+B)}^{-f_c} e^{j2\pi f t} df \\ &= 0.5 \left[\frac{e^{j2\pi(f_c+B)t} - e^{j2\pi f_c t}}{j2\pi t} \right] + 0.5 \left[\frac{e^{-j2\pi f_c t} - e^{-j2\pi(f_c+B)t}}{j2\pi t} \right] \\ &= \frac{0.5}{j2\pi t} \left(e^{j2\pi(f_c+B)t} - e^{j2\pi f_c t} + e^{-j2\pi f_c t} - e^{-j2\pi(f_c+B)t} \right) \\ &= \frac{0.5}{\pi t} \left[\sin(2\pi(f_c+B)t) - \sin(2\pi f_c t) \right]. \end{aligned}$$

$$s_{\text{USB}}(t) = \frac{1}{2\pi t} \left[\sin(2\pi(f_c + B)t) - \sin(2\pi f_c t) \right]$$

There will be a normalization factor 1/2 comparing the two approaches.

2. The ring modulator creates a switching affect by multiplying the input signal $m_1(t)$ with a square pulse train:

$$w_0(t) = \frac{4}{\pi} \left(\cos(2\pi f_c t) - \frac{1}{3} \cos(2\pi \cdot 3f_c t) + \frac{1}{5} \cos(2\pi \cdot 5f_c t) - \dots \right)$$

where $f_c = 100$ Hz. Thus, the input signal to BPF:

$$x_1(t) = w_0(t)m(t) = \frac{4}{\pi} \left(\cos(2\pi f_c t)m(t) - \frac{1}{3} \cos(2\pi \cdot 3f_c t)m(t) + \frac{1}{5} \cos(2\pi \cdot 5f_c t)m(t) - \dots \right)$$

Thus the output at the BPF is:

$$x(t) = \frac{2}{\pi} \cos(2\pi f_c t)m(t)$$

Note down the factor $1/2$.

Another switching modulator, known as the **ring modulator**, is shown in Fig. 4.6a. During the positive half-cycles of the carrier, diodes D_1 and D_3 conduct, and D_2 and D_4 are open. Hence, terminal a is connected to c , and terminal b is connected to d . During the negative half-cycles of the carrier, diodes D_1 and D_3 are open, and D_2 and D_4 are conducting, thus connecting terminal a to d and terminal b to c . Hence, the output is proportional to $m(t)$ during the positive half-cycle and to $-m(t)$ during the negative half-cycle. In effect, $m(t)$ is multiplied by a square pulse train $w_0(t)$, as shown in Fig. 4.6b. The Fourier series for $w_0(t)$ can be found by using the signal $w(t)$ of Eq. (4.5) to yield $w_0(t) = 2w(t) - 1$. Therefore, we can use the Fourier series of $w(t)$ [Eq. (4.5)] to determine the Fourier series of $w_0(t)$ as

$$w_0(t) = \frac{4}{\pi} \left(\cos \omega_c t - \frac{1}{3} \cos 3\omega_c t + \frac{1}{5} \cos 5\omega_c t - \dots \right) \quad (4.7a)$$